Industrial Safety Processes

Electrical Safety

Heart Hat
SAFETY

The Water Analogy

"Electricity flowing through a circuit is similar to the flow of water through a pipe. Although there are a few technical problems with this comparison, it serves to explain the basic function of a standard electrical system."

Heart Hat SAFETY

Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 222.

The Water Analogy

"Voltage (V) is the pressure or push, similar to water pressure forcing the water through a pipe.

The amount of water that is allowed to flow is similar to amperage (I) (the number of electrons moving through the wire).

The pipe will limit or restrict the amount of flow allowed; this is similar to resistance (R) in an electrical conductor (or wire)."

Heart Hat SAFETY

Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 222.

Voltage

- Think of voltage as the pressure in a pipeline.
- Voltage is measured in **volts (v)**.
- Low voltage is any voltage 600V and less.
- Potentially hazardous voltage is between 24 v and 600 v.
- A 12 v car battery in a dead short can be hazardous

Heart Hat
SAFETY

Heart Hat
SAFETY

Resistance

- Think of resistance as a blockage or resistance in a pipe.
- Resistance (R) is any condition that retards current flow.
- Resistance (R) is measured in **ohms (Ω)**
- Resistance generates heat.
- Over-current devices monitor the level of heat in a circuit and stops the current if the heat is too great

Heart Hat
SAFETY

Current

- "…the total volume of electrons moving (like water flowing) past a certain point, in a given length of time."
- Current is measured in amperes (amps)
- The measurement used in relation to electric shock or injury is the milliampere (mA or 0.001 ampere)
- It takes a fraction of an amp to cause death

Heart Hat SAFETY

Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 223.

Watt

- A **watt (W)** is the quantity of electricity that is consumed.
- It is determined by multiplying volts (V) times amps (I).

$$\mathbf{V \times I = W}$$

Volts x Amps = Watts

Heart Hat
SAFETY

Electrical Formulas (CRSTEX / CRSPEX)

Voltage = Current x Resistance

$V = IR$

Current = Voltage ÷ Resistance

$I = V/R$

Resistance = Voltage ÷ Current

$R = V/I$

Heart Hat SAFETY

Current Flow in a Circuit

- If current (I) is to flow through a circuit, a complete path – from the source and back – must be available
- When this path is present, the circuit is said to be complete or “closed”
- When the path is not complete, either intentionally or by accident, the circuit is called “open”

Heart Hat
SAFETY

Short Circuit

- Since electrical current follows the path of least resistance, it would follow this path rather than the one established in the circuit, causing an unusually high flow of current
- Classically, shorts occur when bare wires cross, as for example when the insulation in an old outlet wears away, allowing wires to touch each other

Heart Hat
SAFETY

Main Types of Injuries

- Heart attack
- Exterior burns
- Internal burns – difficult to initially identify
- Knock back – explosion
- Falls from grade or height
- Secondary fires – smoke, burns

Heart Hat
SAFETY

Grounds

- A ground is an object that connects a piece of electrical equipment to earth or some conducting body that serves in place of earth.
- A ground serves to redirect stray electrical currents to the earth in the event of an equipment malfunction.
- NOTE: there should never be electrical currents in the grounds unless something is malfunctioning or damaged

Heart Hat
SAFETY

Grounding

- The purpose of the grounding system is to draw additional current through the grounding path, direct it into the earth, and assist the over-current devices in shutting down the electrical flow.
- **The prevention of human injury or death is <u>NOT</u> the purpose of grounding.**

Heart Hat
SAFETY

Grounding…

- Grounding is designed to detect malfunctions resulting in excess heat and is considered ***FIRE PROTECTION***.
- The over-current device will generally not respond fast enough to prevent injury.
- When protection of people is the concern, other devices (i.e., GFCIs) are used

Heart Hat
SAFETY

Grounding for Static Discharge

- Static electricity may be created by the movement of fluids and other substances through pipes, tubes, chutes, etc.
- Grounding and bonding are important to prevent the build-up and sudden discharge of static electricity
- This is especially important in high hazard locations or working with flammable or explosive materials

Heart Hat
SAFETY

Bonding

- Bonding is the joining of metallic parts to form an electrically conductive path
- Because electrical current will travel from high pressure to low pressure, bonding is designed to keep all of the conductive materials at an equal potential
- This will prevent arcing between conductive objects

Heart Hat
SAFETY

Bonding

- Bonding and grounding are often combined to protect hazardous locations from stray electricity or static build-up
- Bonding is important for activities such as transferring flammable liquids from one container to another

Heart Hat
SAFETY

Fuses & Circuit Breakers

- Devices designed to protect electrical systems from too much power or current flowing through
- Intended to “blow” (fuses) or “trip” (breakers) if there is an overload, wiring problem or short circuit – depending on the design and system
- Blown fuses must be replaced, breakers may be reset

Heart Hat
SAFETY

Portable Electrical Equipment

- Portable electric equipment is common in workplaces and in homes
- Grounding is usually provided through the circuit via a grounding plug
- Equipment may also be double-insulated and may not require ground fault circuit interrupters – though many industrial operations require it

Heart Hat
SAFETY

Two Prong Plugs

- Also known as “polarity plugs”
- The narrow blade is the “hot” connection
- The wide blade is the “neutral” or return connection
- Should be able to plug in device only one way
- Common in small appliances with low power demands

Heart Hat
SAFETY

Grounded Plugs

- Have additional grounding pin to provide some level of protection
- Much more common on all types of tools, lighting, etc.
- Should be minimum standard for all electrical appliances in the workplace
  - Adapters from 2 to 3 prong plugs should not be used in the workplace

Heart Hat
SAFETY

Ground Fault Circuit Interrupter (GFCI)

- A ground fault circuit interrupter (GFCI) is a fast-acting electrical circuit-interrupting device that is sensitive to very low levels of current flow to ground
- The GFCI is designed to sense leaks of current large enough to cause serious personal injury

Heart Hat
SAFETY

GFCIs

- GFCIs only work on line-to-ground fault currents
- GFCIs do NOT give protection to workers in the event of line-to-line contact
- Three main types:
  - Portable
  - Receptacle
  - Circuit Breaker

Heart Hat
SAFETY

Extension Cords

- Should not be used as a “permanent” fix for power – prohibited by Canadian Electrical Code
- Must be part of inspection programs and repaired by qualified person – or discarded – if damaged

Heart Hat
SAFETY

Wire Coloring

- **Black** wire – brass connection
  - Live or “hot” wire in system
- **White** wire – silver connection
  - Neutral or return wire
- **Green** or Bare wire
  - Ground

NOT an exact or standardized system

Heart Hat
SAFETY

Other Key Items

- Electrical installations must be covered at all times
  - There is no exemption for “low voltage”
- Circuits can be tested for faults with simple equipment
- The environment needs to be considered
  - Hazardous locations
  - Potential for mechanical damage
  - Chance of getting wet, encountering water

Heart Hat
SAFETY

Hazardous Location

- These are locations where explosive or flammable gases or vapors, combustible dusts, or ignitable fibers are present or likely to become present.
- These materials can ignite due to electrical causes only if 2 conditions are met:
  - Proper oxygen/substance mix
  - An electrical source generates sufficient heat

Heart Hat
SAFETY

Hazardous Locations

- Special types of electrical installations are required in hazardous locations
- This is to prevent an electrical source of energy to cause ignition
- Type of equipment to be used is based on the type of hazardous location and hazards present
- “Intrinsically safe” or “explosion-proof” are common appliance / tool terms

Heart Hat
SAFETY

Electrical Rooms

- Should may not be used for storage of any times (building & fire codes)
- Should be clean, well-lit, and all high-voltage equipment clearly marked.

Heart Hat
SAFETY

Personal Protective Equipment

- For high voltage installations, special PPE and equipment is required for inspection and maintenance
- New regulations are being implemented in different Canadian jurisdictions and may dramatically affect some workplaces
- Electrically rated clothing is now common – shirts, pants, etc.

Heart Hat
SAFETY

Visual Checks

- Portable tools should be at least visually checked prior to each use
- Items to look for should include:
  - Frayed cords
  - Broken power connections, plugs
  - Damage to housings, casings, covers, etc.
  - Accessibility of on/off switches, triggers, etc.

Heart Hat
SAFETY

System Checks

- It is easy and simple to check for line or ground faults in systems with plug-in devices
- Inspections of industrial workplaces should consider including plug-in tests from time to time due to potential damage to electrical systems

Heart Hat
SAFETY

Lockout / Tagout / Tryout

- Discussed under Safeguarding section
- Very important when working on many systems as electrical energy is the most common energy for most equipment
- Needs to be done right – with locks, not just tags or plastic ties
- Lockout kits and staff training are critical components for safe maintenance

Heart Hat
SAFETY